Started on	Sunday, 24 July 2022, 5:57 PM
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Time taken	6 secs
Marks	0.00/50.00
Grade	**0.00** out of 100.00

Question **1**

Not answered

Marked out of 1.00

Flag question

Which one among the following is the most sedative of tricyclic antidepressants?

Select one:

- Dothiepin
- Nortriptyline
- Lofepramine
- Clomipramine
- Imipramine

Amitriptyline and dothiepin are more sedative, while clomipramine, imipramine, nortriptyline and lofepramine are less sedative.

The correct answer is: Dothiepin

Question **2**

Not answered

Marked out of 1.00

Flag question

Which drug acts as a full agonist at the mu receptor and has a long half-life?

Select one:

- Acamprosate
- Buspirone
- Methadone
- Buprenorphine
- Disulfiram

Explanation:

Methadone is a full agonist at mu opioid receptors and has a long half-life (24 hours).

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 478.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 636.

The correct answer is: Methadone

Question **3**

Not answered

Marked out of 1.00

Flag question

An important mechanism of action for sodium valproate is

Select one:

- Dopamine stabilisation
- NMDA antagonism
- Benzodiazepine antagonism
- GABA potentiation
- Membrane stabilisation

Explanation:

Valproate's mechanism of action is complex and not fully understood. It inhibits the catabolism of GABA (enhancing GABA activity), reduces the turnover of arachidonic acid, activates the extracellular signal-regulated kinase pathway thus altering synaptic plasticity, interferes with intracellular signalling, promotes brain-derived neurotrophic factor expression, and reduces levels of protein kinase C.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 257.

The correct answer is: GABA potentiation

Question **4**

Not answered

Marked out of 1.00

Flag question

The therapeutic effects of conventional antipsychotics is mainly due to

Select one:

- Blockade of dopamine-2 receptors in the mesocortical pathway
- Blockade of alpha-1 receptors
- Blockade of dopamine-2 receptors in the nigrostriatal pathway
- Blockade of dopamine-2 receptors in the tuberoinfundibular pathway
- Blockade of dopamine-2 receptors in the mesolimbic pathway

Explanation:

By the 1970s it was widely recognised that the key pharmacological property of all 'neuroleptics' with antipsychotic properties was their ability to block D2 receptors, specifically those in the mesolimbic/mesostriatal dopamine pathway.

Blockade of dopamine-2 receptors in the mesocortical pathway leads to production or worsening of negative symptoms. Blockade of dopamine-2 receptors in the nigrostriatal pathway leads to EPSEs. Blockade of dopamine-2 receptors in the tuberoinfundibular pathway leads features of hyperprolactinaemia such as galactorrhoea, sexual dysfunction, gynaecomastia, infertility, amenorrhoea, and possibly accelerated osteoporosis. Blockade of alpha-1 receptors leads to sedation, dizziness, orthostatic hypotension.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 162.

The correct answer is: Blockade of dopamine-2 receptors in the mesolimbic pathway

Question **5**

Not answered

Marked out of 1.00

Flag question

Excitotoxicity that is secondary to glutamatergic overstimulation results in neuronal damage. This is the basis of treating patients with neurodegenerative disorders with which of the following medications?

Select one:

- Selegiline
- Donepezil
- Risperidone
- Galantamine
- Memantine

Explanation:

Memantine is a non-competitive NMDA receptor antagonist that may be neuroprotective (protect neurons from glutamate-mediated excitotoxicity). It is licensed for moderate to severe Alzheimer dementia.

Characteristic of cognitive enhancer	Donepezil	Rivastigmine	Galantamine	Memantine
Primary mechanism	AChE-I (selective and reversible)	AChE-I (reversible, non-competitive inhibitor)	AChE-I (competitive and reversible)	NMDA receptor antagonist (non-competitive)
Other mechanism	None	BuChE-I	Nicotine receptor agonist	5-HT3 receptor antagonist
Starting dose	5 mg daily	1.5 mg bd (oral) (or 4.6 mg/24 hours patch)	4 mg bd solution or 8 mg XL daily	5 mg daily
Usual treatment dose (and max dose)	10 mg daily	3-6 mg bd (oral) or 9.5 mg/24 hours patch	8-12 mg bd solution or 16-24 mg XL daily	20 mg daily or (10 mg bd)
Adverse effects	Diarrhoea, nausea, and headache most common	Anorexia, dizziness, nausea, vomiting, and diarrhoea most common	Nausea and vomiting most common	Dizziness, headache, fatigue, diarrhoea, and gastric pain most common
Half-life (hours)	~70 (permitting once daily dosing)	~1 (oral) – active for 10 3.4 (patch)	7-8 (oral solution) 8-10 (XL capsules)	60-100
Metabolism	CYP3A4 CYP2D6 (minor)	Minimal involvement of CYP isoenzymes	CYP3A4 CYP2D6	Primarily non-hepatic
Drug-drug interactions	Yes	Interactions unlikely	Yes	Yes
Effect of food on absorption	None	Delays rate and extent of absorption	Delays rate but not extent of absorption	None
Other	Absorbed completely from the GI tract Fairly well-tolerated and widely used	More likely to cause GI adverse effects than donepezil due to more peripheral activity	More likely to cause GI and neuropsychiatric adverse effects than donepezil	Generally better tolerated than cholinesterase inhibitors

AChE-I = acetylcholinesterase inhibitor
BuChE-I = butyrylcholinesterase inhibitor

Adapted from Maudsley Prescribing Guidelines, 2021. pp. 605-6.

SPMM COURSE

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 605.606.

O'Brien et al. Clinical practice with anti-dementia drugs: a revised (third) consensus statement from the British Association for Psychopharmacology. Journal of Psychopharmacology. 2017 Feb;31(2):147-68.

The correct answer is: Memantine

Question **6**

Not answered

Marked out of 1.00

The chronic administration of tricyclics results in

Select one:

- Up-regulation of beta-adrenergic receptors

Flag question

- Down-regulation of beta and alpha adrenergic receptors
- Up-regulation of alpha-adrenergic receptors
- Down-regulation of alpha-adrenergic receptors
- Down-regulation of beta-adrenergic receptors

Explanation:

It has been consistently reported that chronic administration (2 to 4 weeks of treatment) of antidepressant drugs causes a down-regulation of beta receptors in rats. However, such effects are not observed after acute or single administration of these drugs. Since most of the antidepressant drugs tested thus far produce such effects, it has been suggested that down-regulation or decreased responsiveness of beta-adrenergic receptors may be related to the therapeutic effects manifested by antidepressant drugs. This is yet to be clarified.

In chronic users of tricyclics, the down-regulation of postsynaptic β-adrenergic receptors and their reduced responsiveness may explain hypotensive effects.

Ref: Calvi A, Fischetti I, Verzicco I, Murri MB, Zanetidou S, Volpi R, Coghi P, Tedeschi S, Amore M, Cabassi A. Antidepressant Drugs Effects on Blood Pressure. Frontiers in Cardiovascular Medicine. 2021;8.

The correct answer is: Down-regulation of beta-adrenergic receptors

Question **7**

Not answered

Marked out of 1.00

Flag question

The enzyme/s that metabolise/s most of the body dopamine:

Select one:

- MAO-A and COMT
- COMT only
- MAO-A and MAO-B
- MAO-B and COMT
- MAO-A and Tyrosine hydroxylase

Explanation:

Two enzymes that play significant roles in the degradation of catecholamines are monoamine oxidase and catechol-O-methyltransferase (COMT). Two MAO isoenzymes with differing substrate specificities have been identified: MAO-A, which preferentially deaminates serotonin and norepinephrine, and MAO-B, which deaminates dopamine, histamine, and a broad spectrum of phenylethylamines.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 958.

The correct answer is: MAO-B and COMT

Question **8**

Not answered

Marked out of 1.00

Which one of the following has been postulated as the pharmacological basis of clozapine related hypersalivation?

Select one:

Flag question

Reduced cholinesterase activity

- Activation of the swallowing reflex
- Muscarinic M-4 antagonism
- Adrenergic alpha-2 stimulation
- Muscarinic M-4 agonism

Explanation:

The pharmacological basis of clozapine-related hypersalivation remains unclear. Suggested mechanisms include muscarinic M4 agonism, adrenergic alpha-2 antagonism, and inhibition of the swallowing reflex.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 227.

The correct answer is: Muscarinic M-4 agonism

Question **9**

Not answered

Marked out of 1.00

Flag question

Endogenous substrates for monoamine oxidase (MAO) isoenzymes include

Select one:

- Tyramine, acetylcholine, and dopamine
- Acetylcholine, dopamine, and epinephrine
- Epinephrine, dopamine, and serotonin
- Tyramine, serotonin, and histamine
- Histamine, acetylcholine, and serotonin

Explanation:

Two enzymes that play significant roles in the degradation of catecholamines are monoamine oxidase and catechol-O-methyltransferase (COMT). Two MAO isoenzymes with differing substrate specificities have been identified: MAO-A, which preferentially deaminates serotonin and norepinephrine, and MAO-B, which deaminates dopamine, histamine, and a broad spectrum of phenylethylamines.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 958.

The correct answer is: Epinephrine, dopamine, and serotonin

Question **10**

Not answered

Marked out of 1.00

Flag question

Which of the following increases the risk of withdrawal reactions to therapeutic drugs?

Select one:

- The drug having anticholinergic properties
- Taking the drug after food
- Having multiple active metabolites with slow clearance
- Slow withdrawal of the drug
- Drugs being a partial agonist instead of full agonist

Explanation:

Psychotropics with potent anticholinergic adverse effects may be associated with discontinuation or withdrawal effects.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021.

The correct answer is: The drug having anticholinergic properties

Question **11**

Not answered

Marked out of 1.00

Flag question

How much Zuclopenthixol depot is equivalent to 50mg of haloperidol given every four weeks?

Select one:

- 200mg every 2-4 weeks
- 20 mg every 2-4 weeks
- 20mg every 3 weeks
- 100mg every 2-4 weeks
- 400mg every 2 weeks

Using antipsychotic depot conversion chart from POMH (Prescribing Observatory for Mental Health), 200mg / 2 weeks or 400mg/4 weeks of zuclopentixol depot is equivalent to 50mg of haloperidol depot every 4 weeks.

The correct answer is: 200mg every 2-4 weeks

Question **12**

Not answered

Marked out of 1.00

Flag question

Which of the following is a noradrenaline reuptake inhibitor?

Select one:

- Risperidone
- Reboxetine
- Agomelatine
- Fluoxetine
- Paroxetine

Explanation:

As a noradrenaline reuptake inhibitor, reboxetine is an antidepressant with a novel mode of action. It blocks the norepinephrine reuptake pump (norepinephrine transporter) and presumably, this increases noradrenergic neurotransmission.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 291.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 681.

The correct answer is: Reboxetine

Question **13**

Not answered

Marked out of 1.00

Flag question

Which of the following antidementia drugs acts directly on nicotine receptors?

Select one:

- Donepezil
- Tacrine
- Rivastigmine
- Galantamine
- Memantine

Explanation:

Galantamine, unlike the other anticholinesterases in clinical use, is derived from the alkaloids from the daffodil and snowdrop family. It is a reversible, competitive inhibitor of acetylcholinesterase. It is also an agonist at nicotinic receptor sites. Although a clinically effective drug, galantamine frequently causes gastrointestinal side effects.

Characteristic of cognitive enhancer	Donepezil	Rivastigmine	Galantamine	Memantine
Primary mechanism	AChE-I (selective and reversible)	AChE-I (reversible, non-competitive inhibitor)	AChE-I (competitive and reversible)	NMDA receptor antagonist (non-competitive)
Other mechanism	None	BuChE-I	Nicotine receptor agonist	5-HT3 receptor antagonist
Starting dose	5 mg daily	1.5 mg bd (oral) (or 4.6 mg/24 hours patch)	4 mg bd solution or 8 mg XL daily	5 mg daily
Usual treatment dose (and max dose)	10 mg daily	3-6 mg bd (oral) or 9.5 mg/24 hours patch	8-12 mg bd solution or 16-24 mg XL daily	20 mg daily or (10 mg bd)
Adverse effects	Diarrhoea, nausea, and headache most common	Anorexia, dizziness, nausea, vomiting, and diarrhoea most common	Nausea and vomiting most common	Dizziness, headache, fatigue, diarrhoea, and gastric pain most common
Half-life (hours)	~70 (permitting once daily dosing)	~1 (oral) – active for 10 3.4 (patch)	7-8 (oral solution) 8-10 (XL capsules)	60-100
Metabolism	CYP3A4 CYP2D6 (minor)	Minimal involvement of CYP isoenzymes	CYP3A4 CYP2D6	Primarily non-hepatic
Drug-drug interactions	Yes	Interactions unlikely	Yes	Yes
Effect of food on absorption	None	Delays rate and extent of absorption	Delays rate but not extent of absorption	None
Other	Absorbed completely from the GI tract Fairly well-tolerated and widely used	More likely to cause GI adverse effects than donepezil due to more peripheral activity	More likely to cause GI and neuropsychiatric adverse effects than donepezil	Generally better tolerated than cholinesterase inhibitors

AChE-I = acetylcholinesterase inhibitor
BuChE-I = butyrylcholinesterase inhibitor

Adapted from Maudsley Prescribing Guidelines, 2021. pp. 605-6.

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Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 605-606.

The correct answer is: Galantamine

Question **14**

Not answered

Marked out of 1.00

Flag question

Buspirone acts on

Select one:

- GABA-b receptor complex
- 5HT2a receptor
- 5HT1a and 5HT2a receptor
- 5HT1a receptor

- GABA-a receptor complex

Explanation:

Buspirone is classified as an azapirone (or azaspirone) and is chemically distinct from other psychotropic agents. It is a 5HT1A partial agonist. The potential anxiolytic actions of buspirone could theoretically be due to 5HT1A partial agonist actions at both presynaptic and postsynaptic 5HT1A receptors.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 672.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 370.

The correct answer is: 5HT1a receptor

Question **15**

Not answered

Marked out of 1.00

Flag question

Which of the following drugs is not suitable for treating acute mania?

Select one:

- Topiramate
- Lithium
- Carbamazapine
- Olanzapine
- Valproate

Explanation:

Meta-analysis does not support the use of lamotrigine, gabapentin, or topiramate for acute mania.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 341.

The correct answer is: Topiramate

Question **16**

Not answered

Marked out of 1.00

Flag question

Using antidepressants in combination is generally not advisable. This is especially true when using MAO inhibitors. A drug that is comparatively safe as an adjunctive to MAO inhibitors in treating depression is

Select one:

- Amitriptyline
- Clomipramine
- Fluoxetine
- Fluvoxamine
- Paroxetine

Explanation:

While combination treatments have been generally discouraged, emerging evidence supports combinations to be more effective than monotherapy without increasing drop-outs. Second, the combination of monoamine reuptake inhibitors (selective serotonin reuptake inhibitor, serotonin-norepinephrine reuptake inhibitor, or tricyclic antidepressant) and α2-adrenergic receptor antagonists (mianserin, mirtazapine, trazodone) seems to be the most effective and preferable antidepressant combination.

MAOIs potentiate the actions of TCAs, with some evidence supporting the use of such a combination in the treatment of therapy-resistant depression. One serious interaction between MAOIs and TCAs is the excessive elevation of serotonin. Of the options listed here, amitriptyline has less serotonergic effect than others. Of the TCAs the one with the greatest likelihood for this interaction is clomipramine.

Ref: Henssler J, Alexander D, Schwarzer G, Bschor T, Baethge C. Combining Antidepressants vs Antidepressant Monotherapy for Treatment of Patients With Acute Depression: A Systematic Review and Meta-analysis. JAMA psychiatry. 2022 Feb 16.

The correct answer is: Amitriptyline

Question **17**

Not answered

Marked out of 1.00

Flag question

Which of the following is a 5HT1A partial agonist?

Select one:

- Buspirone
- Pramipexole
- Amisulpride
- Risperidone
- Aripiprazole

Explanation:

Buspirone is classified as an azapirone (or azaspirone) and is chemically distinct from other psychotropic agents. It is a 5HT1A partial agonist. The potential anxiolytic actions of buspirone could theoretically be due to 5HT1A partial agonist actions at both presynaptic and postsynaptic 5HT1A receptors.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 672.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 370.

The correct answer is: Buspirone

Question **18**

Not answered

Marked out of 1.00

Flag question

Which of the following has atypical antipsychotic activity?

Select one:

- Fluphenazine
- Promazine
- Chlorpromazine
- Amoxapine
- Amisulpride

Explanation:

Amisulpride is an atypical, second-generation antipsychotic that has selective and equipotent antagonism for D2 and D3, with negligible affinity for other receptors. Its EPSEs are similar to placebo at lower doses, but it shows dose-dependent EPSEs and prolactinaemia at higher doses.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 212.

The correct answer is: Amisulpride

Question **19**

Not answered

Marked out of 1.00

Flag question

Mechanism of action of acamprosate is

Select one:

- Opioid antagonism
- Glutamate antagonism
- Inhibits alcohol dehydrogenase
- Opioid agonism
- Inhibits aldehyde dehydrogenase

Explanation:

Acamprosate is a synthetic taurine analogue that acts as a functional glutaminergic NMDA antagonist and also increases GABA-ergic function.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 463.

The correct answer is: Glutamate antagonism

Question **20**

Not answered

Marked out of 1.00

Flag question

The drug alprazolam acts on

Select one:

- NMDA receptors
- GABA
- D1 receptors
- Histaminergic receptors
- M1 muscarinic receptors

Explanation:

The benzodiazepines derive their name from their molecular structure. They share a common effect on receptors that have been termed benzodiazepine receptors, which in turn modulate GABA activity. The absorption, the attainment of peak concentrations, and the onset of action are quickest for diazepam, lorazepam, alprazolam, triazolam, and estazolam.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 669.

The correct answer is: GABA

Question **21**

Not answered

Marked out of 1.00

Flag question

The pharmacological activity of benzodiazepines depends on its action on which of the following receptors?

Select one:

- Beta Adrenergic receptor complex
- GABA-a and GABA-b receptor complex
- 5HT1-A receptor complex
- GABA-b receptor complex
- GABA-a receptor complex

Explanation:

There are many different subtypes of GABA-A receptors, depending upon which subunits are present. GABA-A receptors can be categorised into whether they are sensitive to benzodiazepines or insensitive to them. Benzodiazepine-sensitive GABA-A receptors have several structural and functional features that make them distinct from benzodiazepine-insensitive GABA-A receptors.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 258-259.

The correct answer is: GABA-a receptor complex

Question **22**

Not answered

Marked out of 1.00

Flag question

Which of the following statements best describes the receptor profile of trazodone?

Select one:

- It is a serotonin antagonist
- It is a mixed serotonin antagonist/agonist
- It is a serotonin, dopamine and norepinephrine antagonist
- It is a serotonin, dopamine and norepinephrine agonist
- It is a serotonin agonist

Explanation:

The prototype drug that blocks serotonin 2A and 2C receptors as well as serotonin reuptake is trazodone, classified as a serotonin antagonist/reuptake inhibitor (SARI). The SARIs act by potent blockade of 5HT2A receptors as well as dose-dependent blockade of 5HT2C receptors and the serotonin transporter (SERT). Trazodone is a very interesting agent, since it acts like two different drugs, depending upon the dose and the formulation. Trazodone is famous for its effectiveness and utility at low doses as a hypnotic. Higher doses recruit additional known antidepressant receptor actions, including serotonin reuptake inhibition. The active metabolite of trazodone is mCPP, which is an agonist at 5-HT2C receptors.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 311-314.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 639.

The correct answer is: It is a mixed serotonin antagonist/agonist

Question **23**

Not answered

Marked out of 1.00

Flag question

Which one among the following has a stimulant effect?

Select one:

- Dothiepin
- Lofepramine
- Imipramine
- Amitriptyline
- Desipramine

Explanation:

Desipramine has a stimulant effect, and it is the TCA with most evidence for use in ADHD. Studies suggest that, in the short term, desipramine improves the core symptoms of ADHD. However, its effect on the cardiovascular system remains an important clinical concern. Thus, evidence supporting the clinical use of desipramine for the treatment of children with ADHD is low.

Ref: Otasowie J et al. Tricyclic antidepressants for attention deficit hyperactivity disorder (ADHD) in children and adolescents. Cochrane Database of Systematic Reviews. 2014(9).

The correct answer is: Desipramine

Question **24**

Not answered

Marked out of 1.00

Flag question

Which of the following types of drug molecules mimic endogenous chemical transmission?

Select one:

- Antagonists
- Agonists
- Partial agonists
- Non-competitive antagonists
- Irreversible enzyme inducers

Explanation:

An agonist (full agonist) is a drug that binds to a specific receptor producing an effect identical to that usually produced by the neurotransmitter affecting that receptor. Drugs are often designed as receptor agonists to treat a variety of diseases and disorders in which the original neurotransmitter is missing or diminished. A partial agonist is a compound that (even when fully occupying a receptor) possesses affinity for a receptor but elicits a partial pharmacological response at the receptor involved. Partial agonists are often structural analogs of agonist molecules. If neurotransmitter concentrations are low, partial agonists may behave as an agonist. Buprenorphine, buspirone and aripiprazole are all partial agonists.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 593.

The correct answer is: Agonists

Question **25**

Not answered

Marked out of 1.00

Flag question

A patient with Parkinson disease experiences visual hallucinations on levodopa/carbidopa therapy. He is on the lowest dose of levodopa/carbidopa to achieve therapeutic effect and other anti-Parkinsonian medications have not been effective. The most appropriate intervention would be to initiate treatment with which of the following medications?

Select one:

- Aripiprazole
- Amisulpride
- Olanzapine
- Risperidone
- Quetiapine

Explanation:

Low-dose quetiapine is best tolerated in Parkinson disease. Olanzapine, ziprasidone, and aripiprazole are likely to all have greater adverse effects on motor function than quetiapine. Risperidone and typical antipsychotics should be avoided completely.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 815.

The correct answer is: Quetiapine

Question **26**

Not answered

Marked out of 1.00

Flag question

Poor response to lithium maintenance treatment is predicted by

Select one:

- classic mania
- family loading of bipolar spectrum disorders
- rapid cycling
- secondary mania
- obsessional symptoms

Explanation:

Lithium remains the gold standard choice for maintenance treatment in patients with bipolar disorder, especially with a ‘classical’ course of illness. In contrast, rapid cycling is usually defined as bipolar disorder in which four or more episodes of (hypo)mania or depression (or four clear switches in polarity) occur in a 12-month period. It is generally held to be less responsive to drug treatment than non-rapid-cycling bipolar illness.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 344.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 285.

The correct answer is: rapid cycling

Question **27**

Not answered

Marked out of 1.00

Flag question

Drug X produces improvement in depression when given at a dose of 10mg. Drug Y which acts via the same mechanism as X produces the same improvement when given at a dose of 20mg. When X and Y are given together Y binds to more receptors in a quicker and stronger fashion than X. Which of the following is true?

Select one:

- Y has lesser affinity then X
- X is less potent than Y
- X and Y are equally efficacious
- X and Y have equal potency but different affinity
- X and Y have equal affinity but different potency

Efficacy refers to how well the drug produces the expected response i.e. the maximum clinical response produced by a drug (‘productivity'). Efficacy depends on affinity, potency, duration of receptor action in some cases and kinetic properties such as half life, among other factors

The correct answer is: X and Y are equally efficacious

Question **28**

Not answered

Marked out of 1.00

Flag question

Which of the following is a partial D2 agonist?

Select one:

- Risperidone
- Amisulpride
- Pramipexole
- Aripiprazole
- Buspirone

Explanation:

Aripiprazole is a D2 receptor partial agonist. It is also a partial agonist at 5-HT1A receptors, a high affinity antagonist at 5-HT2A receptors, and a low-/moderate affinity antagonist at H1 and alpha-1 receptors. It has no anticholinergic effect.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 212.

The correct answer is: Aripiprazole

Question **29**

Not answered

Marked out of 1.00

Flag

Tyramine gains neuronal access and releases stored catecholamines via which of the following routes?

Select one:

- GABA
- NMDA channel

question

- Calcium channels
- Sodium channels
- Catecholamines reuptake channels

As with most indirect sympathomimetics, tyramine enters the presynaptic neuron through amine uptake pumps - the channels that are blocked by antidepressants such as TCAs.

The correct answer is: Catecholamines reuptake channels

Question **30**

Not answered

Marked out of 1.00

Flag question

Receptor blockade in which dopaminergic pathway results in increased prolactin levels in the body?

Select one:

- Mesocortical pathway
- Nigrostriatal pathway
- Mesolimbic pathway
- Ultra short ganglionic pathway
- Tuberoinfundibular pathway

Explanation:

The dopamine neurons that project from hypothalamus to anterior pituitary gland are known as the tuberoinfundibular dopamine pathway. Normally, these neurons are tonically active and inhibit prolactin release. In the postpartum state, however, the activity of these dopamine neurons is decreased. Prolactin levels can therefore rise during breast feeding so that lactation will occur. If the functioning of the tuberoinfundibular dopamine neurons is disrupted by lesions or drugs, prolactin levels can also rise.

Blockade of dopamine-2 receptors in the mesolimbic pathway leads to the therapeutic effects. Blockade of dopamine-2 receptors in the mesocortical pathway leads to production or worsening of negative symptoms. Blockade of dopamine-2 receptors in the nigrostriatal pathway leads to EPSEs.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 85.

The correct answer is: Tuberoinfundibular pathway

Question **31**

Not answered

Marked out of 1.00

Flag question

Which one of the following has antiepileptic activity?

Select one:

- Fluoxetine
- Buspirone
- Mirtazapine
- Bupropion
- Gabapentin

Explanation:

Gabapentin was first introduced as an antiepileptic drug and was found to have sedative effects that were useful in some psychiatric disorders, especially insomnia. It was also found to be beneficial in reducing neuropathic pain, including postherpetic neuralgia.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 664.

The correct answer is: Gabapentin

Question **32**

Not answered

Marked out of 1.00

Flag question

Which one among the following has high D2/high 5HT2 activity?

Select one:

- Clozapine
- Quetiapine
- Risperidone
- Haloperidol
- Amisulpride

Explanation:

Risperidone is an antagonist of the serotonin 5-HT2A, dopamine D2, alpha-1 adrenergic, alpha-2 adrenergic, and histamine H1 receptors. It is a potent antagonist of D2 receptors. Haloperidol has a high D2/low 5HT2 activity. High D2/high 5HT2 activity is characteristic of atypicals such as olanzapine, risperidone and also loxapine. Low D2/high 5HT2 is specific to clozapine while some argue that quetiapine has a lowD2/low 5HT2 activity.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 609.

The correct answer is: Risperidone

Question **33**

Not answered

Marked out of 1.00

Flag question

Disulfiram acts on

Select one:

- Sigma receptors
- Alcohol dehydrogenase
- Glutamate dehydrogenase
- Aldehyde dehydrogenase
- Nicotinic cholinergic system

Explanation:

Disulfiram causes irreversible inhibition of aldehyde dehydrogenase, resulting in the accumulation of acetaldehyde. This leads to signs and symptoms such as flushing, nausea, vomiting, headache, tachycardia and palpitations if alcohol is consumed. The reaction is dose-dependent; if a large amount of alcohol is consumed, a more severe reaction may lead to hypertension, collapse, and even death. Disulfiram acts as a deterrent (not directly as aversion – in 'aversion', chemicals such as valeric acid or apomorphine are used to create nausea which is paired with alcohol). Disulfiram is most effective in

patients who are supported by family or carers who can supervise disulfiram being taken daily. Disulfiram is no longer contraindicated in patients at risk of psychosis because recommended doses have been lowered. All patients require risk assessment and ongoing monitoring for suicidal behaviour. Patients must be prepared to avoid alcohol in all forms, including mouthwashes and hand gels.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 598.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 465.

The correct answer is: Aldehyde dehydrogenase

Question **34**

Not answered

Marked out of 1.00

Flag question

Which of the following groups of medications are associated with bruxism more often than the others listed?

Select one:

- Stimulants
- Antidementia drugs
- Antidepressants
- Antipsychotics
- Benzodiazepines

Explanation:

Bruxism may occur secondary to the use of psychostimulants.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 502.

The correct answer is: Stimulants

Question **35**

Not answered

Marked out of 1.00

Flag question

The mechanism of action of cyproheptadine is

Select one:

- Histamine H1 receptor blockade
- Serotonin 5HT3 receptor blockade
- Serotonin 5HT2 receptor stimulation
- Histamine H2 receptor blockade
- Histamine H1 receptor stimulation

Explanation:

Cyproheptadine is an antihistamine that is an antagonist of histamine H1 receptors. It is also a 5-HT2 antagonist. Cyproheptadine has been used in the treatment of sexual dysfunction caused by increased serotonin transmission (e.g. SSRIs), particularly anorgasmia. There is also some evidence to support its efficacy in the treatment of akathisia caused by antipsychotics (unlicensed use).

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 178, 640, 925.

The correct answer is: Histamine H1 receptor blockade

Question **36**

Not answered

Marked out of 1.00

Flag question

Weight gain caused by clozapine is possibly related to

Select one:

- Alpha-2 antagonistic activity
- 5HT1-a receptor antagonistic activity
- 5HT2-a receptor antagonistic activity
- 5HT2-c receptor antagonistic activity
- 5HT3 receptor antagonistic activity

Explanation:

Weight gain is a common side effect of antipsychotic medication. Suggested mechanisms include 5-HT2C antagonism, H1 antagonism, D2 antagonism, and increased serum leptin (leading to leptin desensitisation).

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 123.

The correct answer is: 5HT2-c receptor antagonistic activity

Question **37**

Not answered

Marked out of 1.00

Flag question

Which of the following actions can cause anorgasmia?

Select one:

- Beta agonism
- GABA-A agonism
- Alpha 1 agonism
- Antihistaminergic property
- Alpha 1 antagonism

Explanation:

Inhibition of orgasm may be caused by an increase in serotonin activity and raised prolactin, as well as alpha-1 blockade (antagonism).

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 172.

The correct answer is: Alpha 1 antagonism

Question **38**

Not answered

Marked out of 1.00

The mechanism of action of barbiturates is by

Select one:

- Increasing the frequency of chloride channels opening

Flag question

- Increasing the duration of chloride channel opening
- Closing down the potassium channels
- Increasing the number of chloride channels
- Closing down the sodium channels

Barbiturates act on GABA receptors and serve to increase the duration of chloride channel opening.

The correct answer is: Increasing the duration of chloride channel opening

Question **39**

Not answered

Marked out of 1.00

Flag question

A 85-year-old lady with a history of cerebrovascular disease, atrial fibrillation, and prior myocardial infarction presents with moderately severe agitated depression with features of self-neglect. She is taking warfarin, lisinopril, and has been on a maximum dose of venlafaxine for last four weeks with moderate effect. The most appropriate intervention to manage the depression is

Select one:

- Add a benzodiazepine
- Prescribe trazodone
- Add mirtazapine
- Trial of sertraline after stopping venlafaxine
- Electro convulsive therapy

Explanation:

The alpha-2 antagonist mirtazapine is often combined with SNRIs (like venlafaxine) for treatment of cases that do not respond to an SNRI alone. This combination of mirtazapine with an SNRI is sometimes called ‘California rocket fuel’ because of these potentially powerful drugs blasting the patient out of the depths of depression.

Ref: Stahl SM. Stahl’s Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 309.

The correct answer is: Add mirtazapine

Question **40**

Not answered

Marked out of 1.00

Flag question

The mechanism of action of varenicline tartrate is

Select one:

- Nicotinic antagonism
- Partial muscarinic agonism
- Inhibition of nitric oxide synthesis
- Phosphodiesterase inhibition
- Partial nicotinic agonism

Explanation:

Varenicline tartrate is a selective a4ß2 nicotinic acetylcholine receptor partial agonist recommended by NICE to aid with smoking cessation. It can reduce the dopaminergic reward that would occur if a patient did smoke (by competing with nicotine) but also reduce the withdrawal symptoms by stimulating at least some neurotransmission.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 613.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 552.

The correct answer is: Partial nicotinic agonism

Question **41**

Not answered

Marked out of 1.00

Flag question

The endocrine effects of neuroleptics would include all except

Select one:

- Gynaecomastia
- Primary amenorrhoea
- Weight gain
- Lactorrhoea
- Impotence

Explanation:

Hyperprolactinaemia caused by antipsychotics can in turn cause secondary (not primary) amenorrhea in women and breast enlargement and galactorrhea in both men and women. Long-term adverse consequences of hyperprolactinaemia are reductions in bone mineral density and possible increase in the risk of breast cancer. Anticholinergic effects of antipsychotics can cause disorders of sexual arousal, and antipsychotics that block peripheral alpha-1 receptors cause particular problems with erection and ejaculation in men.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 168, 173.

The correct answer is: Primary amenorrhoea

Question **42**

Not answered

Marked out of 1.00

Flag question

Which of the following medications has been found to be effective in patients with premature ejaculation?

Select one:

- Flunarizine
- Fluoxetine
- Flupentixol
- Fluphenazine
- Trifluoperazine

Explanation:

The anorgasmic effects of SSRIs make them useful as a treatment for men with premature ejaculation. Fluoxetine and sertraline have been shown to be effective for this purpose.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 628.

The correct answer is: Fluoxetine

Question **43**

Not answered

Marked out of 1.00

Flag question

A patient on mirtazapine treatment develops rapidly spreading bacterial infection. Which of the following must be immediately checked?

Select one:

- Liver function
- Differential count
- Renal function
- CSF profile
- Blood glucose

Explanation:

Mirtazapine has been associated with leucopenia and, rarely, agranulocytosis.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 914.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 290.

The correct answer is: Differential count

Question **44**

Not answered

Marked out of 1.00

Flag question

A 27-year-old man who is taking mirtazapine for generalized anxiety disorder and major depression experiences significant sedation at a dosage of 7.5 mg/day. Which neurotransmitter receptor is most likely being affected?

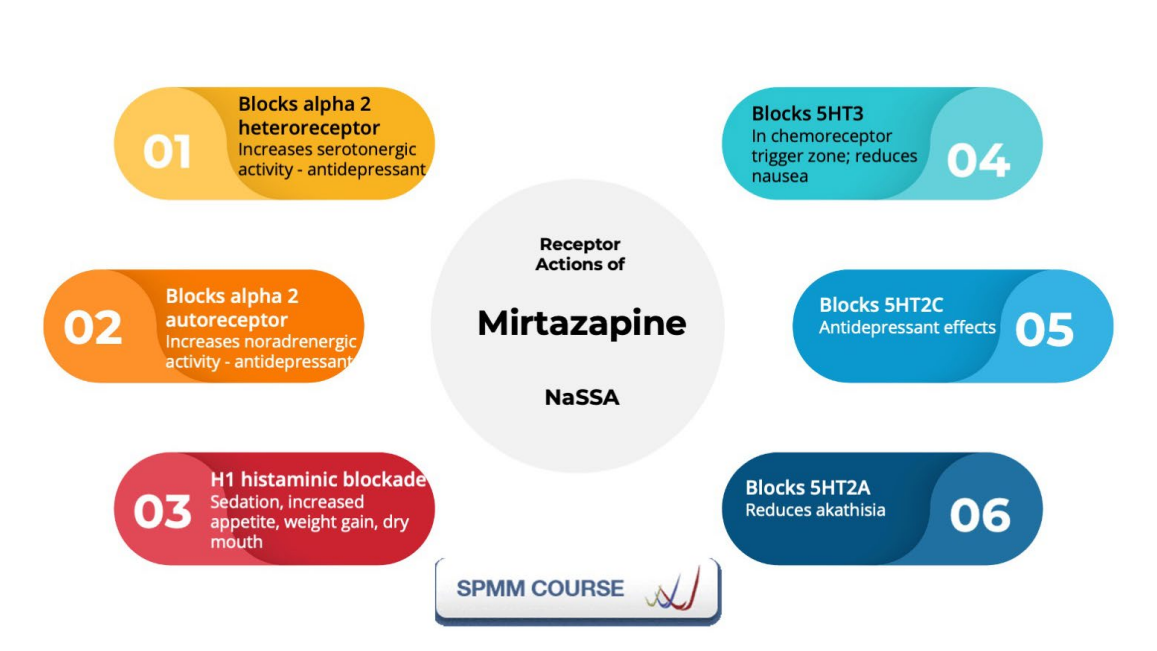
Select one:

- Nicotinic receptors
- Histamine type 1 receptors
- 5HT type 2 receptors
- Muscarinic type 1 receptors
- 5HT type 3 receptors

Explanation:

Both the histaminic and serotonergic effects of mirtazapine contribute to increased sedation. However, the dose of mirtazapine is critical to its effects on sleep. At low doses, mirtazapine preferentially blocks the histamine H1 receptor, since at lower plasma concentrations it has a higher affinity to histamine receptors than to serotonergic receptors. At higher doses, the antihistamine activity is offset by increased noradrenergic transmission, which reduces its sedating effect.

Ref: Leonard SD, Karlamangla A. Dose-dependent sedating and stimulating effects of mirtazapine. Proc UCLA Healthc. 2015;19:1-2.



The correct answer is: Histamine type 1 receptors

Question **45**

Not answered

Marked out of 1.00

Flag question

Mechanism of action of buprenorphine is via partial agonism at

Select one:

- ß-opioid receptors
- µ-opioid receptors
- Adrenergic receptors
- k-opioid receptors
- Cholinergic receptors

Explanation:

Buprenorphine is a synthetic partial agonist with low intrinsic activity and high affinity at mu-opioid receptors.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 483.

The correct answer is: µ-opioid receptors

Question **46**

Not answered

Marked out of 1.00

Flag question

A new drug A is a non-competitive antagonist at NMDA receptors. If availability of glutamate at the receptor sites is increased experimentally after administering drug A, which of the following can happen?

Select one:

- Effect of drug A is partially reversed
- Glutamate levels decrease
- No change takes place

- Effect of drug A is reversed fully
- Effect of drug A increases

Noncompetitive antagonists alter the receptor site in some way so the effects can be reversed only partially by increasing the dose of the agonist drug. Non-competitive antagonism reduces both the potency and the efficacy of agonists.

The correct answer is: Effect of drug A is partially reversed

Question **47**

Not answered

Marked out of 1.00

Flag question

Which one of the following is a predominant D2 antagonist at therapeutic doses?

Select one:

- Aripiprazole
- Clozapine
- Risperidone
- Amisulpride
- Sulpiride

Sulpiride is a predominant D2 antagonist with minimal D3 antagonist/partial agonist actions. Whereas, Amisulpride is a D2/D3 antagonist. At low doses, Amisulpride blocks autoreceptors thus increasing the synaptic dopamine levels. This may explain its therapeutic effects on negative symptoms. In the limbic region, blockade of postsynaptic D2/D3 receptors reduces positive symptoms.

The correct answer is: Sulpiride

Question **48**

Not answered

Marked out of 1.00

Flag question

Which one among the following is a weak inhibitor of cytochrome P450?

Select one:

- Duloxetine
- Fluvoxamine
- Citalopram
- Paroxetine
- Fluoxetine

Explanation:

Citalopram and escitalopram are very weak inhibitors of CYP 1A2 and 2D6. Fluoxetine, fluvoxamine and paroxetine are potent inhibitors of several hepatic cytochrome enzymes. Sertraline is safer, while citalopram is the safest.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 787.

The correct answer is: Citalopram

Question **49**

Not answered

Marked out of 1.00

Flag question

Which of the following street drugs is called a dissociative anaesthetic?

Select one:

- Ketamine
- Cannabis
- Heroin
- LSD
- Cocaine

Explanation:

Ketamine is a compound structurally similar to PCP, used as an anaesthetic and in battlefield surgery. It is a unique anaesthetic, as it does not produce reticular activating system depression, but instead it prevents cortical awareness of painful stimuli. Small amounts can lead to a sense of dissociation. Larger amounts can lead to LSD-like synaesthesia and hallucinations.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 625.

The correct answer is: Ketamine

Question **50**

Not answered

Marked out of 1.00

Flag question

Which of the following statements has no evidence-based support when prescribing the antidementia drug donepezil? It

Select one:

- Reduces caregiver burden
- Improves the activities of daily living in some patients
- Improves neuropsychiatric symptoms
- Reverses the progression of neuropathological degeneration
- Reduces time to institutionalisation

Explanation:

Donepezil may improve symptoms and slow progression of disease but does not reverse the degenerative process. Patients do not generally improve dramatically, although this can be observed in a significant minority of patients. Onset of behavioural problems and nursing home placement can be delayed. Functional outcomes, including activities of daily living, can be preserved. Caregiver burden and stress can be reduced.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 24 246.

The correct answer is: Reverses the progression of neuropathological degeneration

Finish review

PREVIOUS ACTIVITY
Pharmacokinetics

NEXT ACTIVITY
Adverse effects

Jump to...

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